

Comparison of different Kjeldahl Tablets for the Determination of Nitrogen and Protein in Milk according to the Kjeldahl Method



Optimised Kjeldahl analysis with the Büchi KjelMaster K-375

This application note introduces a simple and fast procedure for protein determination in milk according to the Kjeldahl method, as described in the DIN ISO 8968-1:2001, LFBG §64 L01.00-10/1, and AOAC 991.20 regulations.

Comparison in the behaviour of the different types of Kjeldahl Tablets and their individual advantages are summarised here. All the samples described in this application were digested together with sulfuric acid and different Kjeldahl Tablets with the SpeedDigester K-439 and then distilled and titrated with the Kjeldahl sampler system K-375/K-376.

Keywords or phrases: Protein determination, milk, Kjeldahl method, Kjeldahl tablets, Büchi

1. Introduction

An easy and reliable method for determining nitrogen and protein contents in milk according to the Kjeldahl method, as described in the ISO 8968-1:2001, LFBG §64 L01.00-10/1, and AOAC 991.20 regulations, is introduced in chapter 8. All described samples were digested with the SpeedDigester K-436 and K-439. The distillation and boric acid titration are performed with the Kjeldahl sampler system K-375/K-376. In a first step, the new BUCHI Kjeldahl Tablets for performing standard Kjeldahl were compared with each other in order to learn more about the advantages of each type of tablet. For the experiments tryptophan was selected as a reference substance, whole milk and strawberry milk drink were chosen as sample products.

2. Equipment

- › SpeedDigester K-436, K-439 (the parameters used for K-436 are also valid for SpeedDigester K-425)
- › Scrubber K-415 TripleScrub ECO
- › KjelMaster K-375 with KjelSampler K-376
- › Analytical balance (accuracy ± 0.1 mg)

3. Kjeldahl Tablets

Three different types of Kjeldahl Tablets are available for performing standard Kjeldahl. Due to the different weight of the tablets, different volumes of sulfuric acid had to be used during the experiments in order to obtain the optimal ratio of catalyst to sulfuric acid. The composition of each tablet, their weight and the used volume of sulfuric acid are summarized in Table 1.

Weight (g)	Potassium sulfate (%)	Copper sulphate pentahydrate (%)	Titanium (IV) oxide (%)	Sulphuric acid (mL)
3.7	94.3	2.8	2.8	15
4	99.9	0.06	-	16
5	99.6	0.4	-	20

Table 1: Composition of the Kjeldahl Tablets Titanium, ECO and Missouri

4. Chemicals and Materials

- › Sulfuric acid conc. 98 %, Fluka
- › Titanium, BUCHI Kjeldahl Tablet
- › Missouri, BUCHI Kjeldahl Tablet
- › ECO, BUCHI Kjeldahl Tablet
- › Sodium hydroxide 32 %, Brenntag
- › Boric acid 4 %, 200 g boric acid, Brenntag diluted to 5 l with deionized water, pH adjusted to 4.65
- › Sulfuric acid 0.1 mol/l, Fluka standard solution
- › Neutralization solution for the Scrubber: 600 g sodium carbonate, calcined, technical, Synopharm about 2 mL ethanol and a spatula tip of bromothymol blue, Fluka diluted to 3 l with distilled water
- › DL-tryptophan (assay >99 %), Alfa Aesar

5. Samples

- › Whole milk with a labelled protein content of 3.5 g/100 mL

- › Strawberry milk drink with a labelled protein content of 3 g/100 mL

The samples were purchased at a local supermarket.

6. Procedure

The determination of nitrogen and protein in whole milk and strawberry milk includes the following steps:

- › Homogenization of the sample by shaking
- › Digestion of the sample, using SpeedDigester K-439 or K-436
- › Distillation and titration of the sample, using
- › Kjeldahl sampler system K-375/K-376

6.1 Digestion method - tryptophan (verification of the method)

- › Place approx. 0.18 g tryptophan in a 300 mL sample tube
- › Add two tablets of the to be tested catalyst and the corresponding volume of sulfuric acid (98 %) described in Table 1
- › Prepare additional blanks, chemicals without sample
- › Carefully suspend the sample by gently swirling the tube
- › Connect the Scrubber K-415 to the SpeedDigester K-436 or K-439 for absorbing the acid fumes created during digestion
- › Insert the rack containing the samples into the preheated unit
- › Digest the samples according to the parameters listed in Table 2 or 10, depending on the catalyst type.

6.2 Digestion method - samples

- › Place approx. 5 g of the sample in a 300 mL sample tube
- › Add two tablets of the to be tested catalyst and the corresponding volume of sulfuric acid (98 %) described in Table 1
- › Prepare additional blanks, chemicals without sample
- › Carefully suspend the sample by gently swirling the tube

Application Showcase

- Connect the Scrubber K-415 to the SpeedDigester K-436 or K-439 for absorbing the acid fumes created during digestion
- Insert the rack containing the samples into the preheated unit
- Digest the samples according to the parameters listed in Table 2 or 10, depending on the catalyst type.

Step	Temp (°C)	Time (Min)
Preheating	480	-
1	480	10
2	55	10
3	490	Variable ¹
Cooling	-	30

Table 2: Temperature profile for the comparison of the Kjeldahl Tablets with the K-439

¹ In order to be able to determine the progression of the digestion, different digestion times were applied in the third digestion step (490 °C), with varying time periods of 65 min and 95 min. In total the digestion time, of step 1 to 3, was of 85 min and 115 min.

6.3 Distillation and titration

Distil the samples according to the parameters listed in Table 3.

Method parameters KjelMaster K-375			
H ₂ O Volume	60 mL	Titration solution	H ₂ SO ₄ 0.1 mol/L
NaOH volume	90 mL ² / 60 mL ³	Sensor type	Potentiometric
Reaction time	5s	Titration mode	Standard
Distillation mode	Fixed Time	Measuring mode	Endpoint pH
Distillation time	180 s	Endpoint pH	4.65
Stirrer speed distillation	5	Stirrer speed volume	7
Steam output	100%	Titration start volume	0 mL
Titration type	Boric acid	Titration algorithm	Optimal
Receiving solution volume	50 mL		

Table 3: Distillation and titration with the Kjeldahl sampler system K-375/K-376

² Volume of sodium hydroxide used for the comparison of the Kjeldahl Tablets

³ Volume of sodium hydroxide

6.4 Calculation

The results are calculated as a percentage of nitrogen. In order to calculate the protein content of the sample, the nitrogen content is multiplied with a sample-specific protein factor. The following equations (1), (2), and (3) are used to calculate the results. For the reference substance, the purity of the tryptophan is considered in equation (4).

$$w_N = \frac{(V_{\text{Sample}} - V_{\text{Blank}}) \cdot Z \cdot C \cdot f \cdot M_N}{m_{\text{Sample}} \cdot 1000} \quad (1)$$

$$\%N = w_N \cdot 100 \% \quad (2)$$

$$\%P = w_N \cdot PF \cdot 100 \% \quad (3)$$

$$\%N_{\text{Try}} = \frac{\%N \cdot 100}{P} \quad (4)$$

w_N : weight fraction of nitrogen

V_{Sample} : amount of titrant for the sample [mL]

V_{Blank} : mean amount of titrant for the blank [mL]

Z : molar valence factor (1 for HCl, 2 for H₂SO₄)

C : titrant concentration [mol/l]

f : titrant factor (for commercial solutions normally 1.000)

M_N : molecular weight of nitrogen (14.007 g/mol)

m_{Sample} : sample weight [g]

1000 : conversion factor [mL/l]

$\%N$: percentage of weight of nitrogen

$\%P$: percentage of weight of protein

PF : sample-specific protein factor (6.38 for dairy products)

$\%N_{\text{Try}}$: percentage of weight of nitrogen corrected for the purity of reference substance tryptophan (%)

P : purity of the reference substance tryptophan (%)

7. Results for the comparison of the Kjeldahl Tablets

7.1 Recovery of tryptophan

The results of the nitrogen determination and the recovery of tryptophan (assay > 99 %) after 85 and 115

Application Showcase

min, are presented in Table 4-5. The nominal value of tryptophan is 13.72 % of nitrogen.

Tryptophan ^m	m _{sample} [g]	V _{sample} [ml]	%N _{try}	Recovery [%]	Average [%]	RSD [%]
Titanium ^a	0.1733	8.48	13.53	98.6	98.8	0.4
Titanium ^a	0.1858	9.139	13.623	99.3		
Titanium ^a	0.1816	8.872	13.522	98.6		
ECO ^b	0.1741	8.341	13.236	96.5	95.6	1.1
ECO ^b	0.1805	8.581	13.443	95.8		
ECO ^b	0.1939	9.08	12.963	94.5		
Missouri ^c	0.182	8.778	13.335	97.2	96.8	0.3
Missouri ^c	0.1827	8.762	13.259	96.7		
Missouri ^c	0.1786	8.566	13.253	96.6		

Table 4: Results of the recovery of nitrogen in tryptophan with K-439 after 85 min

^a Blank 0.194 mL (n=3)

^b Blank 0.198 mL (n=3)

^c Blank 0.201 mL (n=3)

Tryptophan	m _{sample} [g]	V _{sample} [ml]	%N _{try}	Recovery [%]	Average [%]	RSD [%]
ECO ^b	0.177	8.652	13.522	98.6	98.3	0.4
ECO ^b	0.1822	8.88	13.49	98.3		
ECO ^b	0.1886	9.57	13.448	98		
Missouri ^c	0.1724	8.497	13.617	99.3	99.2	0.3
Missouri ^c	0.1916	9.381	13.558	98.8		
Missouri ^c	0.1891	9.317	13.641	99.4		

Table 5: Results of the recovery of nitrogen in tryptophan with K-439 after 115 min.

7.2 Protein determination in milk

The results of the determination of nitrogen and protein contents in whole milk and strawberry milk drink after 85 and 115 min are presented in Tables 6-9.

Milk drink	m _{sample} [g]	m _{sample} [g]	%N	Protein [%]	Average [%]	RSD [%]
Titanium ^a	5.1331	8.545	0.456	2.91	2.9	0.2
Titanium ^a	5.0626	8.392	0.454	2.89		
Titanium ^a	5.0838	8.45	0.455	2.9		
ECO ^b	5.0963	7.932	0.425	2.71	2.8	3.1
ECO ^b	5.1436	8.432	0.448	2.86		
ECO ^b	5.1437	8.444	0.449	2.87		
Missouri ^c	5.1347	8.467	0.451	2.88	2.9	0.2
Missouri ^c	5.1361	8.46	0.45	2.87		
Missouri ^c	5.1399	8.449	0.45	2.87		

Table 6: Results of the recovery of nitrogen and protein in strawberry milk drink with K-439 after 85 min.

^a Blank 0.086 mL (n=3)

^b Blank 0.090 mL (n=3)

^c Blank 0.098 mL (n=3)

The experimental protein content [%] was re-calculated taking the density of (1.030 g/mL)² into account in order to obtain the protein content as g/100 mL.

Milk drink	m _{sample} [g]	V _{sample} [ml]	%N	Protein [%]	Average [%]	RSD [%]
ECO ^b	5.1212	8.529	0.456	2.91	2.9	0
ECO ^b	5.1004	8.495	0.456	2.91		
ECO ^b	5.1222	8.537	0.456	2.91		
Missouri ^c	5.0566	8.415	0.455	2.9	2.9	0.1
Missouri ^c	5.1903	8.615	0.454	2.9		
Missouri ^c	5.0821	8.459	0.455	2.9		

Table 7: Results of the recovery of nitrogen and protein in strawberry milk drink with K-439 after 115 min.

^b Blank 0.194 mL (n=3)

^c Blank 0.201 mL (n=3)

With the catalyst Titanium and Missouri a digestion time of 85 min is sufficient to have corresponding protein content to the labelling of the strawberry milk drink of 3.0 g/100 mL. To compare the results, the average in % was multiplied with the density of 1.030 g/mL. With the catalyst ECO an elongation of 30 min is necessary to show a complete digestion.

In Table 8 and 9, the results of the determination of nitrogen and protein in whole milk are presented

Whole milk	m _{sample} [g]	V _{sample} [ml]	%N	Protein [%]	Average [%]	RSD [%]
Titanium ^a	5.06	9.768	0.53	3.38	3.4	0.3
Titanium ^a	5.0499	9.783	0.532	3.39		
Titanium ^a	5.0279	9.693	0.529	3.38		
ECO ^b	5.0091	9.557	0.523	3.34	3.3	0.5
ECO ^b	5.0326	9.521	0.519	3.31		
ECO ^b	5.0129	9.563	0.523	3.34		
Missouri ^c	5.0486	9.617	0.522	3.33	3.3	0.1
Missouri ^c	5.0543	9.607	0.521	3.33		
Missouri ^c	5.0223	9.554	0.522	3.33		

Table 8: Results for the recovery of nitrogen and protein in whole milk with K-439 after 85 min

^a Blank 0.194 mL (n=3)

^b Blank 0.198 mL (n=3)

^c Blank 0.201 mL (n=3)

Application Showcase

Whole milk	msample [g]	msample [g]	%N	Protein [%]	Average [%]	RSD [%]
ECO ^b	5.0374	9.688	0.53	3.38	3.4	0.1
ECO ^b	5.0396	9.736	0.53	3.38		
ECO ^b	5.0645	9.803	0.53	3.39		
Missouri ^c	4.9668	9.564	0.53	3.37	3.4	0.3
Missouri ^c	5.0572	9.725	0.53	3.37		
Missouri ^c	5.0429	9.744	0.53	3.38		

Table 9: Results of the recovery of nitrogen and protein in whole milk with K-439 after 115 min.

^b Blank 0.194 mL (n=3)

^c Blank 0.201 mL (n=3)

To compare the results, the average in % was multiplied with the density of 1.030 g/mL.

The labelled protein content for the whole milk is 3.5 g/100 mL. Just with the catalyst Titanium a complete digestion in 85 min is possible. For Missouri and ECO an elongation of 30 min is necessary to reach the labelled protein content of 3.5 g/100 mL.

7.3 Conclusion

With the reference material tryptophan, a good differentiation between the Kjeldahl Tablets is possible. Titanium is the only catalyst, with a recovery (98.8 %; rsd 0.4 %) within the specification of > 98 %, after a digestion time of 85 min. Missouri and ECO need an elongation of 30 min the reach the specification (Figure 1).

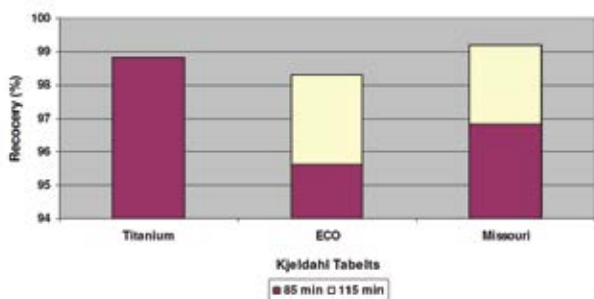


Figure 1: Recovery of tryptophan with different types of Kjeldahl Tablets and digestion times.

In Figure 2 the behaviour of the Kjeldahl Tablets with the sample material strawberry milk drink and whole milk are presented.

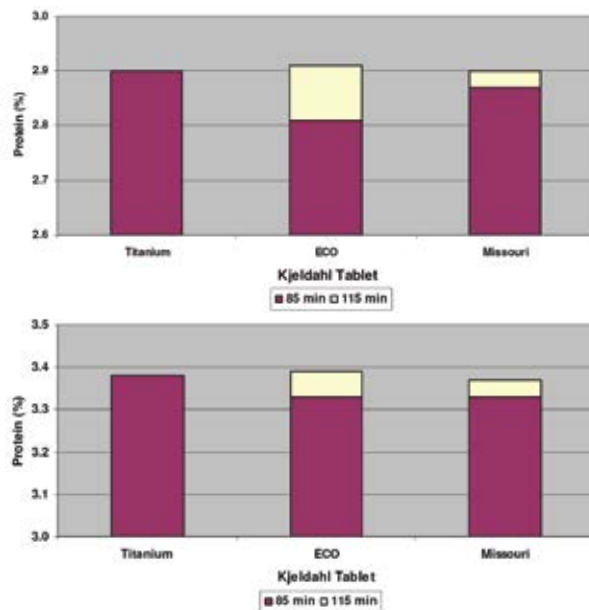


Figure 2: Up: Protein content (%) of strawberry milk drink (n=3) using different Kjeldahl Tablets and digestion times; Below: Protein content (%) of whole milk (n=3) using different Kjeldahl Tablets and digestion times.

The results of the samples are also presented in Table 6-9. The behaviour is similar to the results with the tryptophan. Just the catalyst Titanium is able to digest the samples in 85 min. The catalyst ECO and Missouri need an elongation of 30 min. It is possible, that the combination of copper and titanium helps to digest the ring-bonded nitrogen faster, than the copper alone. In the official regulations¹, a combination of copper sulphate and potassium sulphate, similar to the catalyst Missouri is described.

Kjeldahl Tablet	Total digestion time [min]	Advantage
Titanium	85	Shortest total digestion time
ECO	115	Lowermost copper content
Missouri	115	Similar to AOAC 99120

Table 10: Comparison of the Kjeldahl Tablets

For the digestion with the K-439 and the K-436, the fastest Kjeldahl Tablet, "Titanium", was used for the following application.

8. Results for the digestion conducted with the SpeedDigester K-439 and K-436

The following chapter describes the digestion of the milk samples whole milk and strawberry milk drink which were conducted with the SpeedDigester K-436 and K-439.

The tablet “Titanium” was the only catalyst used for this experiment, the total digestion time was of 85 min.

The preparation of the samples and the calculation of the results are similar to the ones described in Chapter 7. Two tablets “Titanium” and 15 mL of sulfuric acid (98 %) were used (Table 1) for the experiment. The total time of the digestion conducted with the K-439 was of 85 min respective 90 min for K-436. All parameters are specified in Table 11.

	K- 439		K- 436	
Step	Temperature [°C]	Time [min]	Heating Level	Time [min]
Preheating	480	-	8.5	10
1	480	10	8.5	10
2	550	10	9.5	15
3	490	65	8.5	65
Cooling	-	30	-	30

Table 11: Parameters for the digestion with K-439 und K-436

- If the liquid inside the sample tube is not clear and of a blue-green colour after the digestion, digest for an additional 30 min as described in step 3

- Let the samples cool down to room temperature

NOTE: If the samples are placed in the cooling position it takes them approx. 30 min to cool down; if they are left in the heating chamber it takes at least 60 min.

8.1 Digestion with the Kjeldahl Tablet Titanium and K-439

8.1.1 Recovery of tryptophan

The results of the recovery of nitrogen in tryptophan

(assay > 99 %) are presented in Table 12. The nominal value of tryptophan is 13.72 % of nitrogen. The determined values are within the specification of > 98 %¹.

Tryptophan	m _{sample} [g]	V _{sample} [ml]	% N _{Try}	Recovery [%]
Sample 1	0.1733	8.48	13.53	98.6
Sample 2	0.1858	9.139	13.623	99.3
Sample 3	0.1816	8.872	13.522	98.6
Average	-	-	13.56	98.8
Rsd [%]	-	-	0.1	0.1

Table 12: Results of the recovery of nitrogen in tryptophan with K-439.

The mean blank volume for this sample was 0.194 mL (n = 3).

8.1.2 Protein determination in milk

The results of the determination of nitrogen in strawberry milk drink and whole milk are presented in Tables 13 and 14.

Milk drink	m _{sample} [g]	V _{sample} [ml]	% N	% P
Sample 1	5.1331	8.545	0.456	2.91
Sample 2	5.0626	8.392	0.454	2.89
Sample 3	5.0838	8.45	0.455	2.9
Average	-	-	0.455	2.9
Rsd [%]	-	-	0.2	0.2

Table 13: Results of the determination of nitrogen and protein in strawberry milk drink with K-439

The mean blank volume for this sample was 0.194 mL (n = 3).

Whole milk	m _{sample} [g]	V _{sample} [ml]	% N	% P
Sample 1	5.06	9.768	0.53	3.38
Sample 2	5.0499	9.783	0.532	3.39
Sample 3	5.0279	9.693	0.529	3.38
Average	-	-	0.53	3.4
Rsd [%]	-	-	0.3	0.3

Table 14: Results of the determination of nitrogen and protein in whole milk with K-439

Application Showcase

The mean blank volume for this sample was 0.194 mL (n = 3).

The experimental protein content [%] was re-calculated taking the density (1.030 g/mL) into account in order to obtain the protein content as g/100 mL. For the strawberry milk drink, the protein content is 3.0 g/100 mL, for the whole milk 3.5 g/100 mL. The values correspond to the labelled ones, described in chapter 5.

8.2 Digestion with Kjeldahl Tablet Titanium and the K-436

8.2.1 Recovery of tryptophan

The results of the recovery of nitrogen in tryptophan (assay > 99 %) are presented in Table 15. The nominal value of tryptophan is 13.72 % nitrogen. The determined values are within the specification of > 98 %¹

Tryptophan	m sample [g]	V sample [ml]	% N _{Try}	Recovery [%]
Sample 1	0.1755	8.632	13.597	99.1
Sample 2	0.1817	8.964	13.65	99.5
Sample 3	0.1886	9.267	13.605	99.2
Average	-	-	13.62	99.3
Rsd [%]	-	-	0.2	0.2

Table 15: Results of the recovery of nitrogen in tryptophan with K-436

The mean blank volume for this sample was 0.199 mL (n = 3).

8.2.2 Protein determination in milk

The results of the determination of protein in strawberry milk drink and whole milk are presented in Tables 16 and 17.

Milk drink	m sample [g]	V sample [ml]	% N	% P
Sample 1	5.0527	8.43	0.456	2.91
Sample 2	5.2099	8.659	0.455	2.9
Sample 3	4.9259	8.227	0.457	2.91
Average	-	-	0.456	2.91
Rsd [%]	-	-	0	0

Table 16: Results of the determination of nitrogen and protein in strawberry milk drink with K-436.

The mean blank volume for this sample was 0.199 mL (n = 3).

Whole milk	m sample [g]	V sample [ml]	% N	% P
Sample 1	4.9758	9.602	0.529	3.38
Sample 2	5.0644	9.74	0.528	3.37
Sample 3	5.0349	9.756	0.532	3.39
Average	-	-	0.53	3.38
Rsd [%]	-	-	0.4	0.4

Table 17: Results of the determination of nitrogen and protein in whole milk with K-436.

The mean blank volume for this sample was 0.199 mL (n = 3).

The experimental protein content [%] was re-calculated taking the density (1.030 g/mL) into account in order to obtain the protein content as g/100 mL. For the strawberry milk drink, the protein content is 3.0 g/100 mL, for the whole milk 3.5 g/100 mL. These values correspond to the labelled ones, described in chapter 5.

9. Comparison to Standard Methods

The standard methods DIN EN ISO 8968-1:2001 and LFGB §64 L01.00-10/1 are identical. AOAC 991.20 Part "Traditional Method" does not use sucrose and water as blanks. All other parts are identical with DIN EN ISO 8968-1:2001 and LFGB §64 L01.00-10/1. The differences to this application are shown in Table 18.

Application Showcase

	This application	Standard methods	Notes/Impact
Sample tube	300 ml	500-800 ml	No impact
Catalyst	2x 3.7 g Tablets cont. - 47.7 % K ₂ SO ₄ - 2.8 % TiO ₂ - 18 % CuSO	15 g K ₂ SO ₄ + 1ml CuSO ₄ Solution ¹	Easy to handle specially in routine analytics. The choice of catalyst does not influence the result, but the digestion time
Sulfuric acid	15 ml	25 ml	No impact. Same ratio of sulfuric acid / catalyst
Water	60 ml	300-400 ml	The K-375 generates steam in a separated vessel; therefore it is not necessary to add such a high amount of water to the digested sample as described in the standard methods.
Sodium hydroxide	60 ml (Conc.: 32 %)	75 ml (Conc.: 50 %)	No impact. Same ratio of sodium hydroxide /sulfuric acid. Sodium hydroxide 32% is gentler to the pump than higher concentrated alkali.
Blank with sucrose for determination of samples	no	DIN EN ISO and LFGB §64 use 0.85 g and 5 ml water for blank in addition to the chemicals.	No difference observed between blanks with and without sucrose.
Titration	Boric acid titration	AOAC: back titration	No impact. The results are equal with boric acid titration or back titration.

Table 18: Differentiation to the standard methods. ¹ 5 g CuSO₄ x 5 H₂O diluted to 100 mL deionized Water

10. Conclusion

The determination of nitrogen and protein contents in whole milk and strawberry milk drink with the help of the different types of Kjeldahl Tablets provides reliable and reproducible results which correspond to the labelled values of the sample products with low relative standard deviations. The total digestion time can vary between 85 and 115 min depending on the type of Kjeldahl Tablets and digestion unit used and can therefore be adapted to meet the individual needs. The fully-automatic Kjeldahl sampler system, KjelMaster K-375 and KjelSampler K-376, in combination with the unique SpeedDigester models, reduce the time needed for sample analysis significantly.

References

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Operation manual of SpeedDigester K-425 / K-436
Operation manual of SpeedDigester K-439
Operation manual of Scrubber K-415
Operation manual of Kjeldahl sampler system K-375/K-376

